Assignment 8: Manufacturing Analysis

Course: CS101 Computational Thinking | B.Tech 1st Year

Problem Statement

Calculate material and operational costs across production batches. Assess raw material wastage and machine runtime, compute final selling prices based on standardized markups, and identify if the production meets specific profit targets.

Algorithm

1. Create a list of batch tuples: (Batch_ID, Material Cost, Labor Cost, Wastage Percentage, Machine Overhead).
2. Define the minimum acceptable profit margin variable.
3. Iterate over the batches. Calculate total cost encompassing materials, wastage penalty, labor, and machine overhead.
4. Determine the marked-up selling price and the realized profit percentage.
5. Compare realized margin against the target margin and print the status.

Python Implementation

```
# (BatchID, Material_Cost, Labor, Wastage_Pct, Machine_Cost)
production_runs = [
    ("Batch-A", 10000, 3000, 0.05, 2000),
    ("Batch-B", 15000, 4000, 0.12, 3500),
    ("Batch-C", 8000, 2500, 0.02, 1000)
]

target_profit_margin = 15.0 # percentage
markup_multiplier = 1.25    # 25% markup on cost

print("--- Profitability Report ---")
for batch in production_runs:
    b_id, mat_cost, labor, waste_pct, machine = batch

    # Calculate costs
    effective_mat_cost = mat_cost * (1 + waste_pct)
    total_cost = effective_mat_cost + labor + machine

    # Calculate revenue and margins
    selling_price = total_cost * markup_multiplier
    profit = selling_price - total_cost
    margin_percentage = (profit / total_cost) * 100

    # Target check
    status = "FAILED"
    if margin_percentage >= target_profit_margin:
        status = "PASSED"

    print("ID:", b_id)
    print("Cost: Rs.", total_cost, "| Price: Rs.", selling_price)
    print("Margin:", round(margin_percentage, 1), "% -> Target", status)
    print("-" * 30)
```